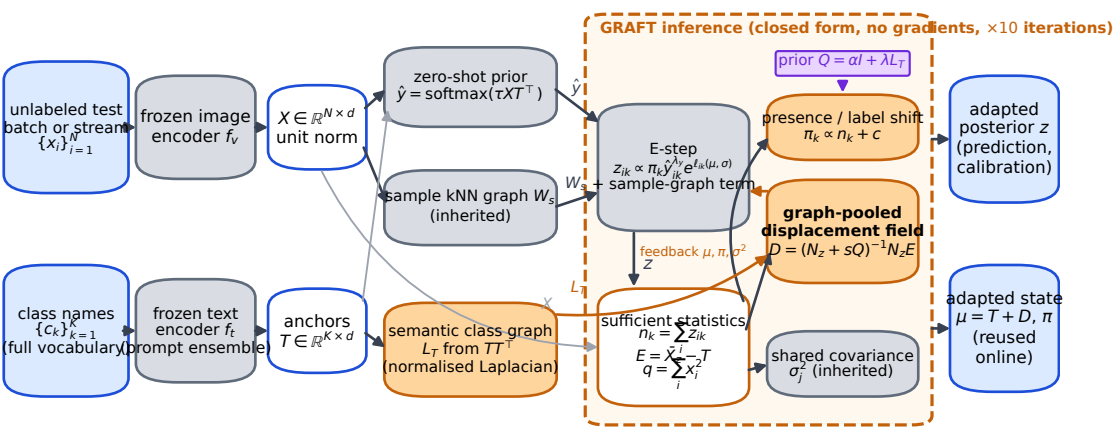
GRAFT inference (closed form, no gradients, ×10 iterations)
unlabeled test batch or stream $\{x_i\}_{i=1}^N$
frozen image encoder f_v
$X \in \mathbb{R}^{N\times d}$ unit norm
zero-shot prior $\hat{y} = \mathrm{softmax}(\tau X T^\top)$
sample kNN graph W_s (inherited)
$\hat{y}$
W_s
E-step $z_{ik} \propto \pi_k \hat{y}_{ik}^{\lambda_y} e^{\ell_{ik}(\mu,\sigma)}$ + sample-graph term
prior $Q = \alpha I + \lambda L_T$
presence / label shift $\pi_k \propto n_k + c$
graph-pooled displacement field $D = (N_z + sQ)^{-1} N_z E$
feedback μ, π, σ^2
Z
adapted posterior z (prediction, calibration)
class names $\{c_k\}_{k=1}^K$ (full vocabulary)
frozen text encoder f_t (prompt ensemble)
anchors $T \in \mathbb{R}^{K\times d}$
semantic class graph L_T from $TT^\top$ (normalised Laplacian)
L_T
Y
sufficient statistics $n_k = \sum_i z_{ik}$ $E = X_i^{-T}$ $q = \sum_i x_i^2$
shared covariance σ_j^2 (inherited)
adapted state $\mu = T + D$, π (reused online)
data / outputs
frozen or inherited
novel (this work)
prior